

Organ-Guided Multi-Phase Abdominal CT Registration via Anatomy-Constrained Mattes Mutual Information

[TODO: Author 1]¹ [TODO: Author 2]¹ [TODO: Author 3]^{1,2}

¹[TODO: Institution]

²[TODO: Institution (if applicable)]

[TODO: corresponding@email.edu]

Abstract

Accurate spatial alignment of multi-phase abdominal computed tomography (CT) — non-contrast (NC), arterial, and venous acquisitions — is essential for joint analysis of vascular enhancement, lesion characterisation, and organ segmentation transfer. The task is complicated by inter-session patient repositioning, variable axial coverage, and the fundamental invalidity of intensity-based similarity metrics across contrast-induced Hounsfield Unit (HU) distributions. We propose a fully automated, anatomy-constrained pipeline comprising five ordered stages: (i) pre-processing with DICOM-convention-aware HU normalisation; (ii) TotalSegmentator-based organ segmentation merged into a custom four-tier label volume; (iii) confidence-gated 3D centroid Z-alignment; (iv) a three-pass rigid registration scheme that initialises from a Kabsch SVD on bone-only labels, refines via Nelder-Mead optimisation of a combined centroid-distance and eroded-Dice loss, and polishes with a Sobel-gradient NCC stage gated by a centroid-displacement acceptance criterion; and (v) B-spline deformable refinement. Throughout, the Mattes Mutual Information (MMI) similarity objective is evaluated exclusively within $\mathcal{M}_{\text{reg}}$, a mask of contrast-stable organs from which bowel, contrast-enhancing structures, and peristaltic tissue are explicitly excluded. Applied to the VinDr multi-phase CT dataset ([265] studies, [795] volumes, two scanner families), the pipeline is evaluated via eroded Dice coefficient and centroid displacement across eleven abdominal structures grouped into parenchymal, vascular, and skeletal categories. **[TODO: Headline result: e.g., “The proposed method achieves a median eroded Dice of $\pm$ across all organ groups, reducing centroid displacement by $\%$ relative to the strongest classical baseline, while requiring $\times$ less wall-clock time than ANTs SyN.”]**

1 Introduction

Multi-phase abdominal CT is a cornerstone of hepatic oncology, vascular surgery planning, and solid organ evaluation [29]. Each protocol comprises three acquisitions — non-contrast (NC), arterial, and venous phases — that capture complementary tissue properties: NC provides baseline parenchymal density, arterial phase reveals vascular enhancement, and venous phase characterises portal and parenchymal perfusion [30]. Joint exploitation of all three phases, whether for lesion characterisation, longitudinal change detection, or segmentation propagation, requires all volumes to be in accurate spatial correspondence.

Achieving this correspondence is non-trivial. First, patients are repositioned between acquisitions, introducing rigid-body translations that include axial shifts up to [80] mm from variable breath-hold depth and table position. Second, contrast administration non-linearly alters the HU distribution of vascular and parenchymal structures: the aortic lumen shifts from ~ 30 HU in NC to ~ 350 HU in arterial phase, while liver parenchyma enhances by 20–80 HU in venous phase [REF]. These shifts violate the local intensity-consistency assumptions of standard similarity metrics such as normalised cross-correlation (NCC) and mean squared error (MSE) [7]. Third, bowel and hollow viscera undergo peristaltic motion and gas redistribution between acquisitions, introducing non-rigid deformations that corrupt any metric evaluated over the full field of view. Prior work has addressed rigid inter-session alignment [REF] and deformable abdominal registration [4, 9] in isolation, yet few methods simultaneously address all three challenges in a single, principled pipeline.

The key gap in existing methods is the application of whole-image similarity objectives in a setting where those objectives are theoretically invalid. Classical toolboxes such as elastix [4] and ANTs [3] default

to full-volume NCC or MSE, metrics whose validity depends on a linear (NCC) or identity (MSE) relationship between corresponding voxel intensities — a relationship broken wherever contrast enhancement occurs. Segmentation-driven approaches [19, 20] use label overlap as the objective, side-stepping intensity invalidity but coupling registration accuracy directly to segmentation quality. A principled formulation is missing: one that retains the statistical power of intensity-based similarity but evaluates it only where the underlying metric assumptions hold.

This paper makes the following contributions:

1. **Anatomy-constrained MMI.** We propose restricting the Mattes Mutual Information objective to $\mathcal{M}_{\text{reg}}$, a TotalSegmentator-derived, automatically merged mask of contrast-stable organs, from which bowel, padding, and contrast-enhancing structures are explicitly excluded. The principled exclusion is motivated by the statistical invalidity of intensity similarity across contrast-induced HU shifts (§4.2; ablated in §6.3).
2. **Bone-anchored Kabsch SVD initialisation.** Prior to intensity optimisation, we align phases via a closed-form Kabsch SVD [27] computed on $\mathcal{M}_{\text{bone}}$, the bone-only sub-mask. Because bony anatomy is both phase-invariant and rigid, this initialisation provides a geometrically sound starting point that is independent of contrast-induced centroid shifts in soft tissue (§4.4).
3. **Confidence-gated 3D centroid Z-alignment.** We introduce a weighted-centroid Z-offset estimation stage with a MAD-filtered confidence score that gates fallback to a centre-of-mass crop when segmentation coverage is insufficient (§4.3).
4. **Organ-grouped evaluation protocol.** We report eroded Dice and centroid displacement per organ, grouped into parenchymal, vascular, and skeletal categories across [265] studies and two contrast phases, providing a reproducible benchmark for future methods on the VinDr dataset (§5).
5. **Systematic comparison and ablation.** We evaluate the full pipeline against **[TODO: ANTs SyN, elastix B-spline, and VoxelMorph]** under identical evaluation conditions, and include an ablation over similarity metric (MMI vs. NCC vs. MSE, masked vs. whole-image) and pipeline stage contribution (§6).

2 Related Work

2.1 Classical Multi-Phase CT Registration

Intensity-based deformable registration was established by Thirion’s Demons algorithm [2] and the free-form B-spline model of Rueckert et al. [1]. The elastix toolkit [4] and ANTs [3] generalised these frameworks to a broad family of similarity metrics and transform models, becoming de facto baselines for abdominal CT registration. Mutual information (MI) [6] and its Parzen-window estimator [5] extended registration to multi-modal settings by modelling the joint intensity histogram rather than assuming linear correspondence; the statistical conditions under which MI remains a valid registration objective are surveyed in [7].

For multi-phase abdominal CT specifically, Metz et al. [9] demonstrated B-spline registration with temporal groupwise optimisation but applied the similarity metric over the full image domain. Heinrich et al. [10] introduced DEEDS, a discrete deformable registration method that optimises over a dense displacement field via efficient message passing, achieving strong performance on abdominal and multi-modal data without explicit intensity assumptions. The Learn2Reg benchmark [11] provides a standardised comparison framework for abdominal registration methods; we adopt a compatible evaluation protocol in §5.

The limitation common to these classical methods is the application of whole-image similarity objectives in the multi-phase setting, where contrast enhancement locally violates the intensity assumptions that justify those objectives.

2.2 Learning-Based Abdominal Registration

VoxelMorph [13] established the amortised learning-based registration paradigm, training a convolutional network to predict deformation fields directly from image pairs. TransMorph [14] and LapIRN [15] subsequently demonstrated strong performance on large-deformation and multi-scale registration. HyperMorph [16] further showed that registration hyperparameters — including the weighting between similarity and regularisation objectives — can themselves be learned, but decisions about *which regions* contribute to the similarity term remain left to the practitioner.

A key challenge for learning-based methods in the multi-phase CT setting is the absence of within-phase training data: most public benchmarks provide intra-

modality pairs, and cross-phase pairs require either manual annotation or a registration method to generate pseudo-ground-truth. UniGradICON [17] represents a step toward task-agnostic generalisation, but its domain shift across contrast phases has not been characterised. The evaluation protocol introduced in §5 is directly applicable to these methods as future baselines.

2.3 Anatomy-Constrained and Masked Registration

The use of anatomical information to guide registration spans several paradigms. Segmentation-driven methods [19,20] replace the intensity objective with a label-based loss, fully decoupling registration from intensity validity but binding accuracy to segmentation quality. Anatomy-constrained neural networks [21] incorporate anatomical shape priors into the training objective, improving plausibility of deformations near organ boundaries.

An alternative approach uses image features that are, by design, modality-independent. Heinrich et al. [22] proposed MIND descriptors that capture local self-similarity structure, making the registration feature representation robust to cross-modal intensity shifts. In the multi-phase CT context, this motivates gradient-domain similarity (used in our Pass 2; §4.4), since HU gradients at organ boundaries are structurally similar across phases even when absolute intensities differ.

Region-of-interest masking within elastix and ITK-based pipelines [4] allows the fixed-image metric domain to be restricted to a spatial mask, but existing applications use this feature as a spatial convenience (e.g., excluding background) rather than as a principled exclusion of contrast-invalid tissue. Sliding-motion regularisation [23] uses anatomical boundaries to adapt the regulariser but evaluates similarity globally. De Vos et al. [24] incorporate anatomy-aware weighting in a learning-based framework, but the weighting is learned from within-modality data and does not address cross-phase invalidity.

The novelty of the present work is the *principled exclusion* of contrast-sensitive and peristaltic structures from an intensity-based objective, motivated by the statistical invalidity of NCC and MSE across contrast-induced HU distributions. To our knowledge, no prior work has formalised this exclusion criterion for multi-phase abdominal CT registration, nor provided an ablation demonstrating its impact relative to whole-image and label-based alternatives.

[TODO: Pipeline diagram: DICOM → NIfTI → TotalSeg → Mask Merge → Crop → Z-align → Rigid (P0/P1/P2) → Deformable → Evaluate. Each box should show the output postfix and the mask it consumes. Distinguish $\mathcal{M}_{\text{reg}}$ (MMI domain) from $\mathcal{M}_{\text{Dice}}$ (Nelder-Mead Dice domain) from $\mathcal{M}_{\text{bone}}$ (Kabsch SVD domain).]

Figure 1: Overview of the proposed multi-phase CT registration pipeline. The NC phase is the fixed reference throughout. Four mask tiers derived from a single TotalSegmentator pass serve distinct roles at each stage.

3 Dataset

We evaluate on the VinDr multi-phase abdominal CT dataset [31], comprising **[265] studies ([795] volumes)** acquired across two scanner families identified by DICOM UID prefixes 1.2.840 and 1.2.392. Each study contains three contrast phases: **non-contrast (NC)**, **arterial**, and **venous**. The NC phase serves as the fixed reference in all experiments; arterial and venous phases are registered independently.

Scanner heterogeneity manifests in differing DICOM encoding conventions: 1.2.840 series require the standard RescaleSlope/Intercept GDCM decoding (which SimpleITK applies automatically), while 1.2.392 series store pixels already in Hounsfield Units. Both conventions are handled identically by the preprocessing stage (§4.1), which clips all volumes to $[-1024, +1024]$ HU after resampling.

[TODO: Report: (1) mean $\pm$ SD slice counts per phase, (2) distribution of axial offsets between NC and moving phases measured at the Z-alignment stage, (3) number of studies successfully processed end-to-end, (4) exclusion criteria and count of excluded studies.]

4 Method

An overview of the pipeline is shown in Fig. 1.

4.1 Preprocessing and Standardisation

Each DICOM series is read via SimpleITK’s ImageSeriesReader (GDCM backend) and processed as follows:

1. **Orientation.** Volumes are reoriented to LPS convention, establishing a canonical axis ordering across scanner families.

2. **Resampling.** Trilinear interpolation resamples to isotropic 1.5 mm spacing. Boundary voxels are filled with -1024 HU (standard CT air), not the scanner minimum, to prevent outside-FOV sentinel values (commonly -4048 HU or -3024 HU) from contaminating the mutual information histogram.
3. **HU clipping.** Values are clipped to $[-1024, +1024]$ HU, removing sentinel artefacts while retaining the full diagnostic range of abdominal soft tissue (≤ 700 HU for cortical bone). This step is applied *before* saving and before TotalSegmentator inference.

4.2 Organ Segmentation and Four-Tier Mask Construction

TotalSegmentator v[**TODO: X.X**] [25] is applied to each preprocessed volume, producing per-structure binary NIfTI files. These are merged into a single multi-label volume using a fixed project-defined integer mapping that assigns each structure a deterministic label independent of TotalSegmentator’s internal indexing. The mapping defines four tiers:

Tier 1 — Bone (labels 21–99): spinal cord, lumbar and lower thoracic vertebrae (L1–L5, T10–T12), sacrum, S1, hip bones, femora, sternum, costal cartilages, and all 24 ribs. These structures are rigid, phase-invariant, and provide the most reliable geometric anchors for registration initialisation.

Tier 2 — Stable soft tissue (labels 1–19): liver, spleen, bilateral kidneys, pancreas, gallbladder, paraspinal muscles (autochthon bilaterally), iliopsoas bilaterally, aorta, inferior vena cava (IVC), portal and splenic vein, and bilateral iliac arteries and veins. These structures maintain predictable spatial positions across contrast phases, making their centroids and intensities (within a known HU range) suitable for metric evaluation.

Tier 3 — Variable soft tissue (labels 100+): stomach, oesophagus, urinary bladder, adrenal glands, prostate, and gluteal muscles. These are included in the bounding-box mask only; they are explicitly excluded from the registration metric domain due to contrast-sensitive HU variation or unpredictable positional change between acquisitions.

Tier 0 — Excluded entirely: colon, small bowel, duodenum, and rectum. Peristaltic motion and gas redistribution between acquisitions introduce non-rigid deformations that are incompatible with any rigid or B-spline model; their inclusion in the metric domain introduces conflicting gradient signals.

Four composite masks are derived from this label volume:

- $\mathcal{M}_{\text{full}}$: union of Tiers 1–3. Used solely to compute the abdominal bounding box for volume cropping; never used as a registration metric domain.
- $\mathcal{M}_{\text{reg}}$: union of Tiers 1–2 (all stable organs). Used as the spatial domain for MMI evaluation in all registration stages, and as the anchor set for Z-alignment centroid estimation.
- $\mathcal{M}_{\text{bone}}$: Tier 1 subset comprising spinal cord, vertebrae L1–L5, T10–T12, sacrum, S1, sternum, and upper ribs (left and right ribs 1–3). Used exclusively in the Pass 0 Kabsch SVD initialisation (§4.4).
- $\mathcal{M}_{\text{Dice}}$: curated Tier 1+2 subset comprising liver, spleen, bilateral kidneys, spinal cord, vertebrae L1–L5, aorta, and IVC. Structures with contrast-variant HU behaviour (iliac vessels) or high peristaltic mobility (stomach, esophagus) are excluded. Used exclusively in the Pass 1 Nelder-Mead Dice loss (§4.4).

4.3 Confidence-Gated Z-Alignment

Inter-session axial offsets are resolved prior to intensity-based registration by weighted centroid matching of anchor organs between phases.

Centroid offset estimation. For anchor organ $k \in \{\text{liver, spleen, kidney}_L, \text{kidney}_R\}$ (labels 1–4 in $\mathcal{M}_{\text{reg}}$), the physical-space Z-centroid $c_k^{(Z)}$ is computed independently for NC and the moving phase. The weighted Z-offset estimate is:

$$\Delta Z = \frac{\sum_{k \in \mathcal{K}} w_k (c_k^{\text{NC},(Z)} - c_k^{\text{mov},(Z)})}{\sum_{k \in \mathcal{K}} w_k} \quad (1)$$

where $\mathcal{K}$ is the set of anchor organs with voxel count ≥ 200 in both phases, $w_{\text{liver}} = 3.0$, and $w_{\text{spleen}} = w_{\text{kidney}_L} = w_{\text{kidney}_R} = 2.0$. Individual organ offsets deviating by more than $1.5 \times$ the median absolute deviation (MAD) from the median are treated as outliers and excluded.

Confidence score and gating. A confidence score $\gamma = \sum_{k \in \mathcal{K}} w_k / \sum_k w_k^{\text{max}}$ reflects the fraction of anchor weight that contributed to the estimate. If $\gamma < 0.40$ or $|\Delta Z| > 80$ mm, the study falls back to a per-series organ centre-of-mass crop. γ , ΔZ , and the method applied are logged per study.

4.4 Three-Pass Organ-Masked Rigid Registration

Rigid registration comprises three sequential passes, each building on the previous.

Pass 0: Kabsch SVD bone initialisation. Given the set of voxel coordinates $\{p_i\}$ from $\mathcal{M}_{\text{bone}}$ in the fixed (NC) phase and $\{q_i\}$ from the corresponding labels in the moving phase, we solve for the optimal rotation R and translation t via the Kabsch algorithm [27]:

$$(R^*, t^*) = \arg \min_{R \in SO(3), t} \sum_i w_i \|R q_i + t - p_i\|^2 \quad (2)$$

where w_i is the per-label weight from $\mathcal{M}_{\text{bone}}$. The closed-form solution factorises the weighted covariance matrix via SVD. Bone labels are chosen for this stage because they are rigid, phase-invariant, and their centroids are unaffected by contrast pooling — properties that do not hold for soft-tissue organ centroids in multi-phase CT. The result $T_0 = (R^*, t^*)$ is used to initialise Pass 1.

Pass 1: Nelder-Mead centroid + eroded-Dice refinement. Starting from T_0 , a derivative-free Nelder-Mead minimiser [28] optimises a combined loss over $\mathcal{M}_{\text{Dice}}$:

$$\mathcal{L}(T) = \lambda \cdot d_c(T) + (1 - \text{Dice}_\epsilon(T)) \quad (3)$$

where d_c is the volume-weighted centroid displacement (Eq. 5), Dice_ϵ is the eroded Dice coefficient (Eq. 4) averaged over $\mathcal{M}_{\text{Dice}}$ labels, and $\lambda = 10.0$ balances the two terms. Nelder-Mead is chosen over gradient-based optimisers because the eroded-Dice term is computed on discretised binary masks and is therefore non-differentiable. The erosion radius per organ is set proportionally to organ size (2.0–6.0 mm), preventing small structures with noisy boundaries from dominating the loss. The result T_1 is used to initialise Pass 2.

Pass 2: Sobel-gradient NCC polish with acceptance gate. Pass 2 applies a phase-invariant intensity polish using SimpleITK’s `ImageRegistrationMethod` with a Sobel-gradient NCC objective: the similarity metric is computed on the gradient-magnitude images $|\nabla I_{\text{fixed}}|$ and $|\nabla I_{\text{moving}}|$ rather than raw HU values. HU gradients at organ boundaries are structurally similar across contrast phases even when absolute intensities differ, providing a phase-invariant intensity signal. Multi-resolution proceeds at shrink factors

[3, 2, 1] with Gaussian smoothing [1.5, 0.5, 0.0] mm, initialised from T_1 . The result T_2 is accepted only when centroid displacement under T_2 is lower than under T_1 ; otherwise T_1 is retained. This acceptance gate prevents the polish stage from corrupting well-converged rigid alignments.

Segmentation masks are propagated to registered space via nearest-neighbour resampling; volumes are resampled with Lanczos windowed sinc interpolation.

4.5 B-Spline Deformable Registration

Deformable refinement uses a free-form B-spline transform [1] at 25 mm control-point grid spacing, applied after the rigid stage. The similarity metric is MMI within $\mathcal{M}_{\text{reg}}$ (fixed mask only; no moving mask is applied, as simultaneous masking introduces conflicting constraints before masks are aligned). Optimisation uses L-BFGS-B with multi-resolution [4, 2, 1] and [X] histogram bins and [Y]% spatial sampling.

[TODO: If grid-spacing ablation (15/25/40 mm) was run, add one sentence referencing §6.2; otherwise remove.]

5 Evaluation Protocol

5.1 Metrics

All metrics are computed between the NC fixed phase and each registered moving phase (arterial, venous), summarised as median $\pm$ IQR. Median and IQR are used throughout as they are robust to the subset of studies where segmentation fails for one phase.

Eroded Dice coefficient. Both fixed and moving organ masks are independently eroded by $r = 6$ mm in physical space prior to Dice computation:

$$\text{Dice}_\epsilon(A, B) = \frac{2|A_\epsilon \cap B_\epsilon|}{|A_\epsilon| + |B_\epsilon|} \quad (4)$$

Erosion removes the boundary voxels where TotalSegmentator uncertainty concentrates, ensuring the metric reflects volumetric alignment quality rather than delineation accuracy. Organ pairs in which either eroded mask contains fewer than 500 voxels are excluded from that study’s statistics.

Centroid displacement (mm). Euclidean distance between physical-space centroids of the full (non-eroded) organ masks:

$$d_c = \|\bar{\mathbf{x}}^{(A)} - \bar{\mathbf{x}}^{(B)}\|_2 \quad (5)$$

where $\bar{\mathbf{x}}^{(\cdot)}$ is the voxel-count-weighted centroid in mm. This metric is insensitive to boundary noise

and sensitive to gross misalignment, making it complementary to eroded Dice.

NCC exclusion. Normalised cross-correlation assumes a linear relationship between corresponding voxel intensities, which is violated across contrast phases [7]. NCC values are noise rather than signal before intensity normalisation and are excluded from all reported results.

5.2 Organ Groups

The eleven evaluated structures are grouped as follows:

Group	Structures
Parenchymal	Liver, Spleen, Kidney (L/R), Pancreas, Gallbladder
Vascular	Aorta, IVC
Skeletal	Vertebra L1, L2, L3

Vertebral bodies are included as contrast-independent internal sanity references: correct rigid registration should yield near-perfect skeletal alignment independent of the similarity metric choice.

5.3 Evaluated Conditions

Pipeline ablation (same framework, varying stages):

Stage	Description
Baseline	Organ CoM crop, no registration
Z-Aligned	Z-centroid alignment only
Rigid	Z-align + three-pass rigid
Deformable	Z-align + rigid + B-spline

Metric ablation (deformable stage, fixed organ mask): MMI (proposed) vs. NCC vs. MSE, each evaluated whole-image and within $\mathcal{M}_{\text{reg}}$. All other hyperparameters held fixed.

Method comparison (deformable stage): [TODO: ANTs SyN [3], elastix B-spline [4], DEEDS [10], VoxelMorph [13]]. Wall-clock time per study (GPU/CPU) reported alongside accuracy.

6 Experiments and Results

6.1 Implementation Details

The pipeline is implemented in Python [3.X] using SimpleITK [v2.X] [26] and TotalSegmentator [vX.X] [25]. Segmentation runs on a single [TODO: GPU model] ([TODO: X] GB VRAM).

Registration (all three passes and B-spline) runs on CPU. [TODO: Report mean $\pm$ SD wall-clock time per study per stage: segmentation / Z-align / rigid (P0 + P1 + P2) / deformable.]

6.2 Pipeline Stage Ablation

Table 1 reports eroded Dice and centroid displacement across all organ groups and phases at each pipeline stage.

[TODO: Analysis paragraph: Which stage provides the largest gain? Does Z-alignment alone close most of the gap, or is intensity registration necessary? Do vascular structures improve less than parenchymal (expected due to smaller volume)? Do vertebrae confirm rigid correctness?]

6.3 Similarity Metric Ablation

Table 2 isolates the contribution of the anatomy constraint from the choice of similarity metric by fixing the pipeline stage (deformable) and varying both the metric and its evaluation domain.

[TODO: Analysis: Show (a) masking consistently improves over whole-image for any metric, and (b) MMI is best within the masked setting. This is the core empirical argument of the paper.]

6.4 Comparison with Existing Methods

[TODO: Analysis: Where does the proposed method outperform baselines most clearly? Any cases where a baseline wins? Discuss the time-accuracy trade-off.]

6.5 Qualitative Results

[TODO: One paragraph referencing Fig. 2. Highlight progression from Baseline to Deformable. Note residual misalignment in difficult structures (pancreas, gallbladder) and explain why.]

7 Discussion

[TODO: D1 — Anatomy-constrained MMI. Argue from Table 2: masking matters more than metric choice (i.e., MMI masked > MMI whole-image $\geq$ NCC masked), or vice versa depending on results. Explain mechanistically: in multi-phase CT, the joint intensity

Table 1: Registration quality at each pipeline stage. Eroded Dice (higher is better) and centroid displacement in mm (lower is better), reported as median $\pm$ IQR across [N] studies per phase. NC is the fixed reference. — = result pending.

Stage	Ph.	Parenchymal		Vascular		Skeletal	
		Dice $\uparrow$	CD $\downarrow$	Dice $\uparrow$	CD $\downarrow$	Dice $\uparrow$	CD $\downarrow$
Baseline	A	—	—	—	—	—	—
	V	—	—	—	—	—	—
Z-Aligned	A	—	—	—	—	—	—
	V	—	—	—	—	—	—
Rigid	A	—	—	—	—	—	—
	V	—	—	—	—	—	—
Deformable	A	—	—	—	—	—	—
	V	—	—	—	—	—	—

[TODO: Figure 2: Axial overlay panels for a representative study. Show NC (fixed) and arterial (moving) at four stages: Baseline / Z-Aligned / Rigid / Deformable. Use checkerboard or edge overlay. Annotate liver, aorta, L2. Show one easy and one hard (large Z-offset) case.]

Figure 2: Qualitative registration results for a representative study (axial mid-liver slice). NC fixed phase shown in green, arterial moving phase in magenta. Misalignment appears as colour fringing; correct alignment produces a grey overlay.

Table 2: Effect of similarity metric and masking on deformable registration quality. All conditions use identical pipeline stages and hyperparameters. Median eroded Dice / centroid displacement (mm). — = result pending.

Condition	Arterial		Venous	
	Dice $\uparrow$	CD $\downarrow$	Dice $\uparrow$	CD $\downarrow$
MSE (whole image)	—	—	—	—
NCC (whole image)	—	—	—	—
MSE (organ mask)	—	—	—	—
NCC (organ mask)	—	—	—	—
MMI (whole image)	—	—	—	—
MMI (organ mask)	—	—	—	—

histogram between NC and arterial liver contains two distinct modes (enhancing vs. non-enhancing parenchyma). Whole-image MMI marginalises over this heterogeneity; organ-masked MMI evaluates it where tissue identity is known.]

[TODO: D2 — Kabsch SVD initialisation. Argue that Pass 0 provides a geometrically correct warm start that prevents Pass 1 from converging to local optima in the

Dice/centroid loss surface. If the Pass 0 ablation is available (rigid with and without Kabsch init), quantify the gain here.]

[TODO: D3 — Organ-group observations. Parenchymal: liver and spleen show the largest absolute improvement. Pancreas and gallbladder: small volume, contrast-variable, expect higher variance. Vascular: aorta is rigid and contrast-bright in arterial; expect good alignment. IVC is compressible — may show higher centroid displacement. Skeletal: any residual centroid error here indicates a registration failure, not an anatomy challenge.]

[TODO: D4 — Limitations. (1) No deep learning component — iterative inference. Report total study time; discuss suitability for retrospective cohort analysis vs. real-time. (2) Pipeline depends on TotalSegmentator quality; segmentation failures propagate downstream (characterise failure rate). (3) Evaluation uses pseudo ground truth (TotalSegmentator masks); no manually annotated landmarks. (4) Extension to other datasets (CHAOS, LiTS, multi-centre trials) requires verifying DICOM encoding conventions.]

Table 3: Comparison against existing registration methods at the deformable/non-rigid stage on [265] studies. CD = centroid displacement (mm). Time = mean wall-clock seconds per study (CPU unless noted). **Bold** = best per column. — = result pending.

Method	Parenchymal		Vascular		Skeletal		Time (s)
	Dice ↑	CD ↓	Dice ↑	CD ↓	Dice ↑	CD ↓	
ANTs SyN [3]	—	—	—	—	—	—	—
elastix B-spline [4]	—	—	—	—	—	—	—
DEEDS [10]	—	—	—	—	—	—	—
VoxelMorph [13]	—	—	—	—	—	—	—
Proposed	—	—	—	—	—	—	—

8 Conclusion

We presented a multi-phase abdominal CT registration pipeline whose central design principle is the confinement of intensity similarity to an automatically derived mask of contrast-stable organs. By evaluating Mattes Mutual Information exclusively within $\mathcal{M}_{\text{reg}}$ — a mask from which bowel, contrast-enhancing vascular tissue, and peristaltic structures are explicitly excluded — the pipeline avoids the corrupting influence of phase-dependent HU variation on the registration objective. A bone-anchored Kabsch SVD initialisation and a derivative-free Nelder-Mead refinement stage, together with a Sobel-gradient NCC acceptance gate, provide geometrically robust convergence across scanner types and large axial offsets. **[TODO: Quantitative summary: e.g., “On [265] VinDr studies, the proposed method achieves —% higher median eroded Dice and —mm lower centroid displacement than the strongest classical baseline.”]** The four-tier mask construction and eroded-Dice evaluation protocol provide a reproducible benchmark for future registration methods on multi-phase abdominal CT.

References

- [1] D. Rueckert *et al.*, “Nonrigid registration using free-form deformations: Application to breast MR images,” *IEEE Trans. Med. Imag.*, vol. 18, no. 8, pp. 712–721, 1999.
- [2] J.-P. Thirion, “Image matching as a diffusion process: An analogy with Maxwell’s demons,” *Med. Image Anal.*, vol. 2, no. 3, pp. 243–260, 1998.
- [3] B. B. Avants *et al.*, “Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain,” *Med. Image Anal.*, vol. 12, no. 1, pp. 26–41, 2008.
- [4] S. Klein *et al.*, “Elastix: A toolbox for intensity-based medical image registration,” *IEEE Trans. Med. Imag.*, vol. 29, no. 1, pp. 196–205, 2010.
- [5] D. Mattes *et al.*, “Nonrigid multimodality image registration,” in *Proc. SPIE Med. Imag.*, vol. 4322, 2001, pp. 1609–1620.
- [6] W. M. Wells III *et al.*, “Multi-modal volume registration by maximization of mutual information,” *Med. Image Anal.*, vol. 1, no. 1, pp. 35–51, 1996.
- [7] J. P. W. Pluim, J. B. A. Maintz, and M. A. Viergever, “Mutual-information based registration of medical images: A survey,” *IEEE Trans. Med. Imag.*, vol. 22, no. 8, pp. 986–1004, 2003.
- [8] C. Studholme, D. L. G. Hill, and D. J. Hawkes, “An overlap invariant entropy measure of 3D medical image alignment,” *Pattern Recognition*, vol. 32, no. 1, pp. 71–86, 1999.
- [9] C. T. Metz *et al.*, “Nonrigid registration of dynamic medical imaging data using nD+t B-splines and a groupwise optimization approach,” *Med. Image Anal.*, vol. 15, no. 2, pp. 238–249, 2011.
- [10] M. P. Heinrich *et al.*, “DEEDS: MRF-based non-linear image registration using efficient discrete optimization,” *Int. J. Comput. Vis.*, vol. 103, no. 3, pp. 384–400, 2013.
- [11] A. Hering *et al.*, “Learn2Reg: Comprehensive multi-task medical image registration challenge, dataset and evaluation,” *IEEE Trans. Med. Imag.*, vol. 41, no. 6, pp. 1458–1476, 2022.

- [12] K. Murphy *et al.*, “Semi-automatic construction of reference standards for evaluation of image registration,” *Med. Image Anal.*, vol. 15, no. 1, pp. 71–84, 2011.
- [13] G. Balakrishnan *et al.*, “VoxelMorph: A learning framework for deformable medical image registration,” *IEEE Trans. Med. Imag.*, vol. 38, no. 8, pp. 1788–1800, 2019.
- [14] J. Chen *et al.*, “TransMorph: Transformer for unsupervised medical image registration,” *Med. Image Anal.*, vol. 82, p. 102615, 2022.
- [15] T. C. W. Mok and A. Chung, “Large deformation diffeomorphic image registration with Laplacian pyramid networks,” in *Proc. MICCAI*, 2020, pp. 211–221.
- [16] A. Hoopes *et al.*, “HyperMorph: Amortised hyperparameter learning for image registration,” in *Proc. IPMI*, 2021, pp. 3–17.
- [17] H. Tian *et al.*, “UniGradICON: A foundation model for medical image registration,” *arXiv:2403.05780*, 2024.
- [18] Z. Zhao *et al.*, “Recursive cascaded networks for unsupervised medical image registration,” in *Proc. ICCV*, 2019, pp. 10600–10610.
- [19] Z. Xu and M. Niethammer, “DeepAtlas: Joint semi-supervised learning of image registration and segmentation,” in *Proc. MICCAI*, 2019, pp. 420–429.
- [20] Y. Hu *et al.*, “Weakly-supervised convolutional neural networks for multimodal image registration,” *Med. Image Anal.*, vol. 49, pp. 1–13, 2018.
- [21] O. Oktay *et al.*, “Anatomically constrained neural networks (ACNN): Application to cardiac image enhancement and segmentation,” *IEEE Trans. Med. Imag.*, vol. 37, no. 2, pp. 384–395, 2018.
- [22] M. P. Heinrich *et al.*, “MIND: Modality independent neighbourhood descriptor for multi-modal deformable registration,” *Med. Image Anal.*, vol. 16, no. 7, pp. 1423–1435, 2012.
- [23] B. W. Papiez *et al.*, “An implicit sliding-motion preserving regularisation via bilateral filtering for deformable image registration,” *Med. Image Anal.*, vol. 18, no. 8, pp. 1323–1335, 2014.
- [24] B. D. de Vos *et al.*, “A deep learning framework for unsupervised affine and deformable image registration,” *Med. Image Anal.*, vol. 52, pp. 128–143, 2019.
- [25] J. Wasserthal *et al.*, “TotalSegmentator: Robust segmentation of 104 anatomical structures in CT images,” *Radiol.: Artif. Intell.*, vol. 5, no. 5, 2023.
- [26] B. C. Lowekamp *et al.*, “The design of SimpleITK,” *Front. Neuroinform.*, vol. 7, p. 45, 2013.
- [27] W. Kabsch, “A solution for the best rotation to relate two sets of vectors,” *Acta Crystallogr. A*, vol. 32, no. 5, pp. 922–923, 1976.
- [28] J. A. Nelder and R. Mead, “A simplex method for function minimization,” *Comput. J.*, vol. 7, no. 4, pp. 308–313, 1965.
- [29] A. Forner, M. Reig, and J. Bruix, “Hepatocellular carcinoma,” *Lancet*, vol. 391, no. 10127, pp. 1301–1314, 2018.
- [30] European Association for the Study of the Liver, “EASL clinical practice guidelines: Management of hepatocellular carcinoma,” *J. Hepatol.*, vol. 69, no. 1, pp. 182–236, 2018.
- [31] **[TODO: VinDr multi-phase CT dataset citation. Search PhysioNet or authors’ publications for the correct reference. Placeholder.]**